# Concepte fundamentale ale limbajelor de programare

## Tipuri de date abstracte

## Curs 09

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# Cuprins


1. Tip stiva de intregi
2. Modul stiva de intregi
3. Tip stiva de tip_el
4. Modul stiva de tip_el

# Tip stiva de intregi

```
type stack(nr_max:integer)=class

 operations
  pop, push, top;

 var tab_st:array[1..nr_max] of integer;
 ind:integer;
```

# Tip stiva de intregi

```
function pop():integer;
begin
 if not empty() then
  ind:=ind-1;
  return tab_st[ind];
 else
  exception;
 end if;
end;
```

# Tip stiva de intregi

```
procedure push(x:integer);
begin
 if not overflow() then
  tab_st[ind]:=x;
  ind:=ind+1;
 else
  exception;
 end if;
end;
```

# Tip stiva de intregi

```
function top():integer;
begin
 if not empty() then
  return tab_st[ind-1];
 else
  exception;
 end if;
end;
```

# Tip stiva de intregi

```
function empty():boolean;
begin
 return ind=1;
end;

function overflow():boolean;
begin
 return ind>nr_max;
end;
```

# Tip stiva de intregi

```
begin
 ind:=1;
end;
```

# Exemplu de utilizare

```
var
 st1,st2 : stack(100);
 st3 : stack(55);

st1.push(10);
...
st2.push(k);
...
i:=st1.top();
...
j:=st2.pop();
```

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# Modul stiva de intregi

```
module st;

export
 stack,pop,push,top,init;

type stack(nr_max:integer)=record
 tab_st:array[1..nr_max] of integer;
 ind:integer;
end record;
```

# Modul stiva de intregi

```
function pop(var s:stack):integer;
begin
 if not empty(s) then
  s.ind:=s.ind-1;
  return s.tab_st[s.ind];
 else
  exception;
 end if;
end;
```

# Modul stiva de intregi

```
procedure push(var s:stack;x:integer);
begin
 if not overflow(s) then
  s.tab_st[s.ind]:=x;
  s.ind:=s.ind+1;
 else
  exception;
 end if;
end;
```

# Modul stiva de intregi

```
function top(s:stack):integer;
begin
 if not empty(s) then
return s.tab_st[s.ind-1]
 else
  exception;
 end if;
end;
```

# Modul stiva de intregi

```
procedure init(var s:stack);
begin
 s.ind:=1;
end;

function empty(var s:stack):boolean;
begin
 return s.ind=1;
end;
```

# Modul stiva de intregi

```
function overflow(var s:stack):boolean;
begin
 return s.ind>s.nr_max;
end;

var
 st1,st2:st.stack(100);
 st3:st.stack(50);
```

# Exemplu de utilizare

```
st.init(st1);
st.init(st2);
st.init(st3);
...
st.push(st1,10);
...
st.push(st2,k);
...
i:=st.top(st1);
...
j:=st.pop(st2);
```

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4. Modul stiva de tip_el

# Tip stiva de tip_el

```
type stack(type tip_el, nr.max:integer)=class

operations
 pop, push, top;

var var.st:array[1..nr.max]of tip.real;
ind:integer;
```

# Tip stiva de tip_el

```
function pop():tip_el;
begin
...
end;

procedure push(x:tip_el);
begin
...
end;
```

# Tip stiva de tip_el

```
function top():tip_el;
begin
...
end;

function empty():boolean;
begin
...
end;
```

# Tip stiva de tip_el

```
function overflow():boolean;
begin
...
end;

begin
...
end;
```

# Exemplu de utilizare

```
type t=--------
var
 st1 : stack(integer,100);
 st2 : stack(real,150);
 st3 : stack(t,50);
...
st1.push(10);
st2.push(1.5);
```

# Cuprins


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4. Modul stiva de tip_el

# Modul stiva de tip_el

```
module st(type tip_el);

export
 stack, pop, push, top, init;

type stack(nr.max:integer) = record
 tab_st:array[1..nr.max] of tip_el;
 ind:integer;
 end record;
```

# Modul stiva de tip_el

```
function pop(var s:stack) : tip.el;
begin
...
end;

procedure push(var s:stack; x:tip.el);
begin
...
end;
```

# Modul stiva de tip_el

```
function top(var s:stack) : tip.el;
begin
...
end;

procedure init(var s:stack);
begin
...
end;
```

# Modul stiva de tip_el

```
function empty(var s:stack) : boolean;
begin
...
end;

function overflow(var s:stack) : boolean;
begin
...
end;

begin
...
end;
```

# Exemplu de utilizare

```
type t = ------------;
module st_int = st(integer);
module st_real = st(real);
module st_t = st(t);
```

```
var
 st1 : st_int_stack(100);
 st2 : st_real_stack(150);
 st3 : st_t_stack(50);
...
st_int.push(st1,10);
...
st_real.push(st2,1.5);
...
```

# Bibliografie


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3. Horia Ciocarlie – Universul limbajelor de programare, editia 2-a, editura Orizonturi Universitare, Timisoara, 2013.